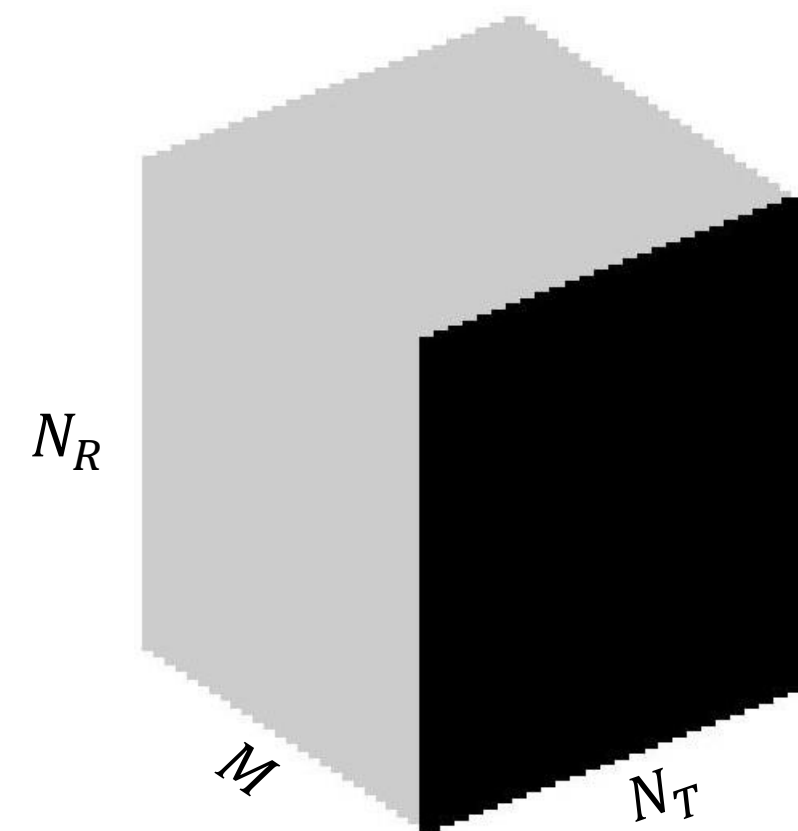
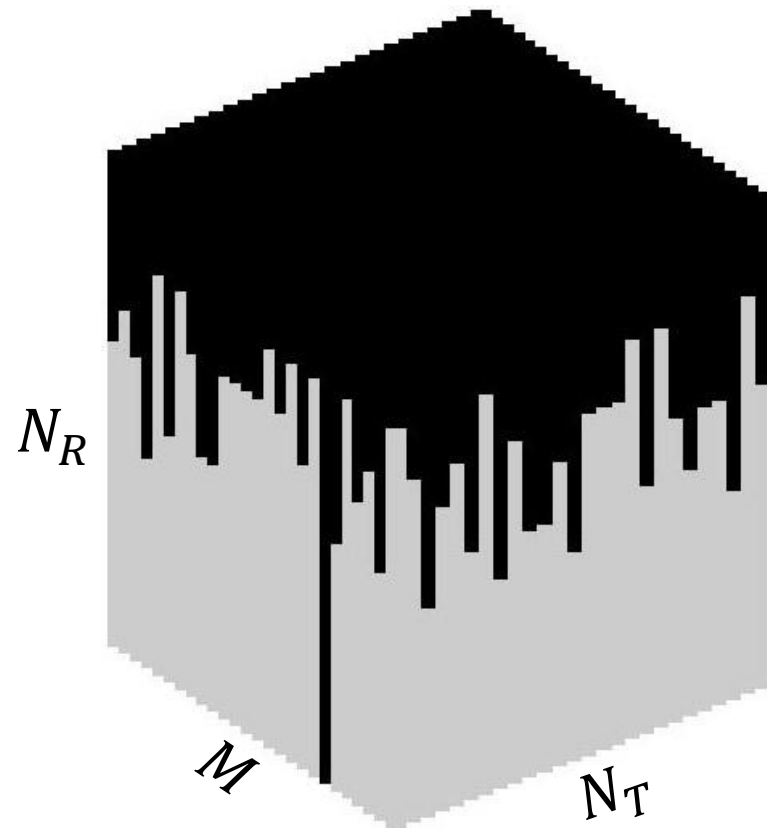
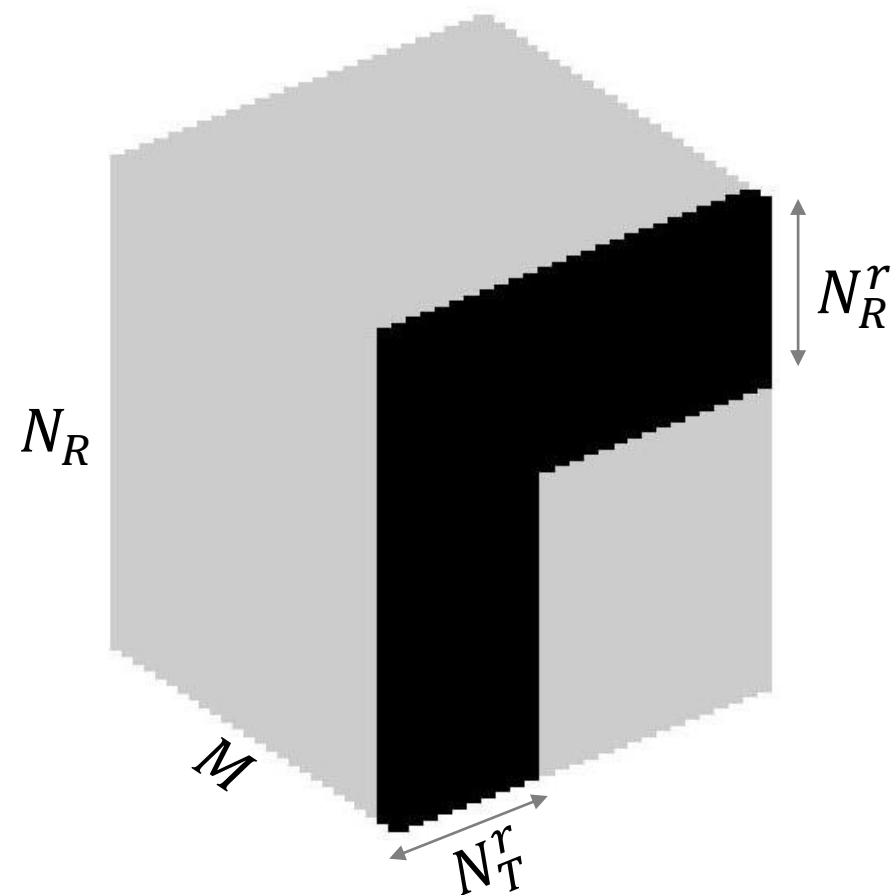




(a) The HL-ML and the MRC Detectors – by avoiding the computations along the M dimension – incur an order of complexity of $N_R N_T$ instead of $M N_R N_T$ required for the direct ML approach



(b) The m-M ML Detector reduces the complexity by avoiding some computations along the N_R dimension



(c) The Reduced Rank MRC Detection lower the complexity count of the HL-ML and the MRC detectors approximately by $(N_T - N_T^r)(N_R - N_R^r)$